

TRONDHEIM VAERNES, Norway

WMO: 012710

Lat: 63.459N

Lon: 10.935E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.4	7.6	-3.9	4.5	4.6	0.8	5.7	10.8	36.7	35.8	31.0	35.6	7.9	120	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	77.3	62.3	73.7	61.3	70.2	59.5	64.7	73.5	62.8	71.1	60.9	68.4	9.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.0	80.4	68.9	59.1	75.0	66.8	57.3	70.1	64.4	29.7	73.6	28.3	71.2	27.0	68.9	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 27.9	(b) 23.7	(c) 20.4	(e) -1.1	(f) 82.9	(g) 5.7	(h) 4.0	(i) -5.2	(j) 85.8	(k) -8.5	(l) 88.1	(m) -11.7	(n) 90.4	(o) -15.9	(p) 93.3		
(a) 27.9	(b) 23.7	(c) 20.4	(e) -1.7	(f) 67.5	(g) 5.5	(h) 2.6	(i) -5.7	(j) 69.4	(k) -8.9	(l) 70.9	(m) -12.1	(n) 72.4	(o) -16.1	(p) 74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	29.3	29.6	33.4	41.3	48.5	54.3	59.6	58.5	52.3	43.0	34.7	30.3
	DBStd	13.23	9.41	8.99	7.31	6.10	6.73	5.74	5.89	5.59	5.73	6.63	8.40	9.66
	HDD50	3474	643	572	515	270	111	18	1	2	39	232	460	611
	HDD65	8082	1108	992	980	711	513	323	189	216	383	682	909	1076
	CDD50	909	0	0	0	9	64	148	300	265	107	15	1	0
	CDD65	44	0	0	0	0	2	4	23	14	1	0	0	0
	CDH74	302	0	0	0	0	11	28	169	92	2	0	0	0
	CDH80	49	0	0	0	0	0	2	35	12	0	0	0	0
	WSAvg	8.7	9.5	9.3	9.3	8.5	8.7	9.0	8.1	7.7	8.1	8.2	8.5	9.0
	PrecAvg	39.8	3.0	3.0	3.1	2.3	2.5	3.7	3.9	3.8	4.1	3.6	3.2	3.6
(15) Precipitation	PrecMax	48.2	7.8	6.4	7.4	5.2	3.8	5.8	6.9	6.6	9.6	6.8	8.9	7.6
	PrecMin	26.2	0.0	0.3	1.1	0.8	1.3	1.4	1.9	1.6	1.5	1.4	0.4	0.7
	PrecStd	5.2	1.8	1.5	1.6	1.1	0.7	1.1	1.4	1.4	2.2	1.2	2.3	1.8
	PrecStd	5.2	1.8	1.5	1.6	1.1	0.7	1.1	1.4	1.4	2.2	1.2	2.3	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	46.7	47.8	51.0	65.4	75.4	77.8	84.7	81.4	72.1	64.2	55.2	49.5	49.5
	0.4% MCWB	40.0	41.4	42.3	50.8	57.1	60.1	66.0	62.7	58.5	53.5	48.1	42.9	42.9
	2% DB	44.4	44.9	47.5	58.9	69.4	73.0	78.4	76.4	68.0	59.3	51.1	46.4	46.4
	2% MCWB	39.2	39.4	40.8	47.5	55.7	59.7	63.7	62.9	57.1	50.6	45.2	40.8	40.8
	5% DB	42.6	43.0	45.0	54.9	64.3	68.7	74.4	72.2	64.5	56.1	48.3	44.3	44.3
	5% MCWB	37.8	38.2	39.3	45.7	53.4	57.5	62.6	61.3	55.4	48.9	42.7	39.5	39.5
	10% DB	40.7	40.8	43.0	51.4	60.4	64.8	70.5	68.6	61.3	52.7	46.0	42.2	42.2
	10% MCWB	36.6	36.8	38.1	43.8	50.7	56.0	60.9	59.6	53.9	47.0	41.2	38.0	38.0
	0.4% WB	42.2	42.9	44.1	51.7	59.9	63.5	68.8	66.3	61.1	54.7	49.2	44.7	44.7
	0.4% MCDB	44.0	45.5	47.9	62.4	69.9	73.1	80.4	75.8	68.9	61.8	52.8	47.4	47.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.3	40.7	42.2	48.6	56.8	60.7	65.7	64.1	58.4	51.7	46.2	42.0	42.0
	2% MCDB	43.0	43.7	45.9	56.9	67.3	70.1	75.0	72.6	65.2	57.8	50.3	44.9	44.9
	5% WB	38.5	38.7	40.3	46.5	54.1	58.6	63.7	62.3	56.6	49.7	43.4	40.1	40.1
	5% MCDB	41.5	41.8	44.2	52.8	63.5	66.7	71.9	70.4	62.4	55.0	47.0	43.2	43.2
	10% WB	36.8	37.0	38.7	44.5	51.4	56.6	61.6	60.5	54.9	47.7	41.4	38.3	38.3
	10% MCDB	39.8	40.2	42.1	50.5	59.1	64.0	69.2	67.3	60.2	52.3	45.3	41.3	41.3
(35) Mean Daily Temperature Range	MDBR	9.6	10.1	11.8	13.3	14.2	13.2	14.0	13.9	12.7	11.0	9.4	9.9	9.9
	5% DB	9.6	10.1	12.2	18.8	21.0	21.0	21.5	20.4	17.4	14.3	11.7	10.2	10.2
	5% WB	7.0	7.1	8.2	10.7	11.1	11.0	10.3	10.3	9.3	8.8	8.2	7.7	7.7
	MCDBR	9.2	9.4	11.3	16.6	20.0	18.4	19.9	18.2	15.4	12.8	11.2	9.7	9.7
	MCWBR	7.3	7.5	8.3	10.1	11.1	10.3	10.4	10.0	9.3	8.6	8.6	7.9	7.9
	MCWBR	7.3	7.5	8.3	10.1	11.1	10.3	10.4	10.0	9.3	8.6	8.6	7.9	7.9
(40) Clear-Sky Solar Irradiance	taub	0.229	0.290	0.322	0.350	0.354	0.354	0.363	0.350	0.329	0.307	0.366	0.357	0.357
	taud	2.035	2.210	2.251	2.341	2.418	2.457	2.454	2.484	2.531	2.436	2.684	2.611	2.611
	Ebn at Noon	119	201	243	263	273	275	269	262	246	200	114	64	64
	Edh at Noon	9	19	27	31	32	31	30	27	21	16	8	5	5
(44) All-Sky Solar Radiation	RadAvg	49	216	613	1179	1446	1508	1466	1125	675	299	78	23	23
	RadStd	8	33	89	95	98	155	151	132	70	40	13	2	2

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page